in comparison to that of every other star, makes it by far the dominant influence on the earth's motion. And so, we can get a ballpark estimate by considering only the sun's gravitational influence. For many purposes this is perfectly adequate. If necessary, we can refine this approximation by sequentially including the gravitational effects of the next-most-relevant bodies, such as the moon and whichever planets are passing closest by at the moment. The calculations can start to become difficult as the emerging web of gravitational influences gets complicated, but don't let this obscure the perturbative philosophy: The sun-earth gravitational interaction gives us an approximate explanation of the earth's motion, while the remaining complex of other gravitational influences offers a sequence of ever smaller refinements.

A perturbative approach works in this example because there is a dominant physical influence that admits a relatively simple theoretical description. This is not always the case. For example, if we are interested in the motion of three comparable-mass stars orbiting one another in a trinary system, there is no single gravitational relationship whose influence dwarfs the others. Correspondingly, there is no single dominant interaction that provides a ballpark estimate, with the other effects yielding small refinements. If we tried to use a perturbative approach by, say, singling out the gravitational attraction between two stars and using it to determine our ballpark approximation, we would quickly find that our approach had failed. Our calculations would reveal that the "refinement" to the predicted motion arising from the inclusion of the third star is not small, but in fact is as significant as the supposed ballpark approximation. This is familiar: The motion of three people dancing the hora bears little resemblance to two people dancing the tango. A large refinement means that the initial approximation was way off the mark and the whole scheme was built on a house of cards. You should note that it is not simply a matter of including the large refinement due to the third star. There is a domino effect: The large refinement has a significant impact on the motion of the other two stars, which in turn has a large impact on the motion of the third star, which then has a substantial impact on the other two, and so on. All strands in the gravitational web are equally important and must be dealt with simultaneously. Oftentimes, in such cases, our only recourse